

Yuhao Yuan

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EDUCATION

University of Toronto	Sept 2025 – Now
Master of Education, Ends in Jul 2027	
Beijing Normal University	Sept 2016 – Jul 2021
Bachelor of Computer Science	

WORK EXPERIENCE

Software Developer - ByteDance Ltd. China	June 2021 - Jul 2024
- Supported Continuous Integration and Continuous Delivery of basic components for China Tiktok iOS Application. - Worked on solving user needs and making the app easier to use in different languages and regions. - Developed two new applications as the main developer.	
FrontEnd Software Developer Internship - Alibaba Ltd. China	Jun 2020 - Sept 2020
Used a fast technique called SIMD (Single Instruction, Multiple Data) to apply a blur effect and smooth out the pictures.	

ACADEMIC EXPERIENCE

Researcher - Dr. GPT	Jan 2026 - Present
- Maintained the English and Chinese versions of an LLM-powered psychological assessment platform. - Built CI/CD pipelines to automate build and deployment workflows. - Contributed to platform deployment and service delivery in mainland China.	
Researcher - pyKeyATM project	Jan 2025 - Now
Collaborated with Dr. Ji at the University of Toronto to Develop the pyKeyATM R package, designed for abstracting topic words from documents.	
Researcher - rBKT Project	Aug 2024 - Dec 2025
- Developed an open-source R implementation of Bayesian Knowledge Tracing for estimating student knowledge mastery from problem-solving sequences. - Contributed to model training and prediction APIs, data validation, simulation, evaluation, parameter-recovery experiments, and package documentation.	
Researcher - Fed IRT	Jan 2024 - Aug 2024
- Co-developed FedIRT, an open-source R package for privacy-preserving estimation of 1PL, 2PL, and graded response models using federated learning. - Implemented and evaluated FedAvg workflows and a Shiny client-server interface, enabling institutions to estimate IRT parameters collaboratively without sharing raw data.	

Reasercher - Entrepreneurial Project

Aug 2021 - Jul 2024

- Developed an Android App called “keysManager” using KotlirLanguage
- Launch the product outside of school, continuously tracked the users’ online activities, and up-dated the App according to the target market’s feedback
- Accomplished the goal of the App to manage user’s passwords,create secure passwords, set up network Keychain, and automatically sync all data while maintainingnetwork security

Researcher - Graduation Design

Aug 2021 - Jul 2024

- Generating 3D model texture maps from actual images.
- An algorithm using convex hull nearest matching points to iteratively refine the projection matrix.
- Introduced an automated process for generating texture maps from multiple images, significantly reducing the need for manually marked initial correspondence points.

Researcher - Team Performance Evaluation Model in Soccer

Feb 2020 - Apr 2020

- Collaborated on developing a comprehensive model to evaluate soccer team performance using network analysis of player interactions.
- Led the team and won the first prize.

PUBLICATIONS

- **Yuan, Y.**, Zhou, B., Qi, J., Luo, N., & Ji, F. (2026). BKT: A Bayesian knowledge tracing package for the R environment. *Behavior Research Methods*, 58(5), Article 134. doi:10.3758/s13428-026-02955-9
- Zhou, B., **Yuan, Y.**, Luo, N., Qi, J., & Ji, F. (2025). FedIRT: An R package and Shiny app for estimating federated item response theory models. *Journal of Open Source Software*, 10(113), 7869. doi:10.21105/joss.07869

AWARDS

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| • National Competition Medalist Scholarship | 2021 |
| • MCM/ICM (Mathematic-Modeling) Finalist | 2020 |
| • MCM/ICM (Mathematic-Modeling) Meritorious | 2018 |
| • International Collegiate Programming Contest (Shenyang) Silver Meda | 2019 |
| • International Collegiate Programming Contest (Jiaozuo) Bronze MTedal | 2018 |
| • International Collegiate Programming Contest (Beijing) Bronze Medal | 2017 |
| • China Collegiate Computing Contest 3rd Place (Group) | 2019 |
| • China Collegiate Programming Contest (Harbin) Silver Medal | 2019 |
| • China Collegiate Programming Contest (Qinhuangdao) Bronze Meddal | 2018 |
| • BNU Programming Contest (7th annual contest) 2nd Place | 2018 |
| • BNU Programming Contest (6th annual contest) 2nd Prize | 2017 |
| • Jingshi Academic Excellence Scholarship | 2017 |